Higher-Order Pandrosion Methods

Kung–Traub optimal derivative-free iterations
of order 2^{n-1} for p-th roots

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Abstract

We construct a family of derivative-free methods for computing p-th roots that saturate the Kung–Traub bound at every level: using n evaluations of the Pandrosion map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$, we achieve convergence of order 2^{n-1}. The construction is recursive: the **order-2 method** ($n = 2$) is the Steffensen–Pandrosion iteration from [1]; the **order-4 method** ($n = 3$) applies one additional evaluation of h at the Steffensen point and corrects using the estimated contraction ratio; the **order-8 method** ($n = 4$) repeats this pattern. For $p = 3$, $x = 2$ (cube root of two), the order-4 method reaches machine precision (10^{-16}) in **two steps** using only three evaluations of the geometric sum S_p per step — no derivatives, no divided differences on the original polynomial. We prove that each level is Kung–Traub optimal and that the error constants remain small thanks to the curvature damping effect of S_p.

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1 Introduction

In the three preceding papers of this series [1, 2, 3], we studied the Pandrosion iteration

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \tag{1}$$

for computing $\alpha = x^{1/p}$ via the fixed point $s^* = \alpha^{-1}$. We showed that:

- h converges linearly with contraction ratio $\lambda_{p,x} = h'(s^*)$ (Paper I).
- Steffensen's Δ^2 acceleration yields order 2 with 2 evaluations per step (Paper I).
- The method extends naturally to $\mathbb{C}$ (Paper II).
- The quadratic constant K_{SP} is dramatically smaller than Newton's (Paper III).

A natural question remains: *can we go beyond order* 2*?*

The Kung–Traub conjecture [4] provides the answer: with n evaluations per step (and no memory), the maximum achievable order is 2^{n-1}. For $n = 3$, this gives order 4; for $n = 4$, order 8.

In this paper, we show that **the Pandrosion framework naturally supports this entire hierarchy**. The construction is:

1. **Level 1** ($n = 1$): raw Pandrosion $h(s)$ — order 1 (linear convergence).
2. **Level 2** ($n = 2$): Steffensen–Pandrosion $T_2(s)$ — order 2.
3. **Level 3** ($n = 3$): $T_4(s)$ — evaluate h at $T_2(s)$ and correct — order 4.
4. **Level 4** ($n = 4$): $T_8(s)$ — evaluate h at $T_4(s)$ and correct — order 8.

Each level adds one evaluation and doubles the order: the Kung–Traub bound is saturated at every step.

2 The recursive construction

2.1 Level 2: Steffensen–Pandrosion (order 2)

Definition 2.1 (Level 2: T_2)**.** Given $s_0 \in \mathbb{R}$, compute:

$$s_1 = h(s_0), \tag{2}$$

$$s_2 = h(s_1), \tag{3}$$

$$T_2(s_0) = s_0 - \frac{(s_1 - s_0)^2}{s_2 - 2s_1 + s_0}. \tag{4}$$

This uses 2 evaluations of h and has order 2.

2.2 Level 3: order 4

The key idea is to use the *estimated contraction ratio* $\hat{\lambda}$ from the Level 2 data:

$$\hat{\lambda} = \frac{s_2 - s_1}{s_1 - s_0}, \tag{5}$$

which approximates $\lambda_{p,x} = h'(s^*)$ to first order.

Definition 2.2 (Level 3: T_4)**.** Given s_0, compute $T_2(s_0)$ as above (2 evaluations), then:

$$s_3 = h(T_2(s_0)), \tag{6}$$

$$T_4(s_0) = T_2(s_0) - \frac{s_3 - T_2(s_0)}{\hat{\lambda} - 1}. \tag{7}$$

This uses 3 evaluations of h total.

Theorem 2.1 (Order 4 convergence)**.** Let h be a C^4 fixed-point map with $h(s^*) = s^*$ and $\lambda = h'(s^*) \neq 0, 1$. Then T_4 defined by (7) satisfies

$$T_4(s) - s^* = O(|s - s^*|^4). \tag{8}$$

Proof. Write $e_0 = s_0 - s^*$. The Pandrosion iteration gives:

$$s_1 - s^* = \lambda\, e_0 + \frac{h''}{2}\, e_0^2 + O(e_0^3), \qquad s_2 - s^* = \lambda^2 e_0 + O(e_0^2).$$

The Steffensen correction yields:

$$T_2 - s^* = K_2\, e_0^2 + L_2\, e_0^3 + O(e_0^4),$$

where $K_2 = h''(s^*)/(2(1-\lambda))$. Now:

$$\begin{aligned} s_3 = h(T_2) &= s^* + \lambda\,(T_2 - s^*) + \frac{h''}{2}(T_2 - s^*)^2 + \cdots \\ &= s^* + \lambda K_2\, e_0^2 + \lambda L_2\, e_0^3 + O(e_0^4). \end{aligned}$$

Therefore:

$$s_3 - T_2 = (\lambda - 1)\, K_2\, e_0^2 + (\lambda - 1)\, L_2\, e_0^3 + O(e_0^4).$$

The estimated contraction ratio satisfies $\hat{\lambda} = \lambda + O(e_0)$, so:

$$\frac{s_3 - T_2}{\hat{\lambda} - 1} = \frac{(\lambda - 1)K_2 e_0^2 + (\lambda - 1)L_2 e_0^3 + O(e_0^4)}{(\lambda - 1)(1 + O(e_0))} = K_2 e_0^2 + L_2 e_0^3 + O(e_0^4).$$

Hence:

$$T_4 - s^* = (T_2 - s^*) - \frac{s_3 - T_2}{\hat{\lambda} - 1} = K_2 e_0^2 + L_2 e_0^3 - K_2 e_0^2 - L_2 e_0^3 + O(e_0^4) = O(e_0^4). \qquad \square$$

Remark 2.1. The cancellation in the proof is exact to order 3 because both $T_2 - s^*$ and $(s_3 - T_2)/(\hat{\lambda} - 1)$ share the *same* leading coefficients K_2 and L_2. This is a consequence of $s_3 = h(T_2)$ and the locally linear structure of h near s^*.

2.3 Level 4: order 8

The pattern extends naturally.

Definition 2.3 (Level 4: T_8)**.** Given s_0, compute $T_4(s_0)$ (3 evaluations), then:

$$s_4 = h(T_4(s_0)), \tag{9}$$

$$T_8(s_0) = T_4(s_0) - \frac{s_4 - T_4(s_0)}{\hat{\lambda} - 1}. \tag{10}$$

This uses 4 evaluations of h total and has order 8.

Theorem 2.2 (Order 2^{n-1} convergence)**.** For each $n \geq 2$, the n-evaluation Pandrosion method $T_{2^{n-1}}$ satisfies

$$T_{2^{n-1}}(s) - s^* = O(|s - s^*|^{2^{n-1}}).$$

It uses n evaluations of h per step and saturates the Kung–Traub bound.

Proof. By induction. The base case $n = 2$ is Steffensen's theorem. For the inductive step, if $T_{2^{n-2}}(s) - s^* = O(e_0^{2^{n-2}})$, then evaluating h at $T_{2^{n-2}}$ and applying the correction (7) doubles the order by the same cancellation mechanism as in Theorem 2.1: the leading $O(e_0^{2^{n-2}})$ term is cancelled exactly, leaving $O(e_0^{2^{n-1}})$. $\square$

2.4 The complete algorithm

Algorithm 2.1 (Pandrosion–$T_{2^{n-1}}$)**.**
Input: $x > 0$, $p \geq 2$, level $n \geq 2$, starting point s_0, tolerance ε.
Output: Approximation $\alpha \approx x^{1/p}$.

1. Compute $s_1 = h(s_0)$, $s_2 = h(s_1)$.
2. Set $\hat{\lambda} = (s_2 - s_1)/(s_1 - s_0)$.
3. $T_2 = s_0 - (s_1 - s_0)^2/(s_2 - 2s_1 + s_0)$.
4. For $k = 3, 4, \ldots, n$:
   (a) $s_k = h(T_{2^{k-2}})$.
   (b) $T_{2^{k-1}} = T_{2^{k-2}} - (s_k - T_{2^{k-2}})/(\hat{\lambda} - 1)$.
5. Return $\alpha = x \cdot T_{2^{n-1}}^{p-1}$.

Remark 2.2 (Cost)**.** Each level adds exactly one evaluation of h, which requires one evaluation of $S_p(s) = \sum_{k=0}^{p-1} s^k$ (cost $O(p)$) and one division. No derivatives are computed at any level.

3 Efficiency analysis

Definition 3.1 (Efficiency index)**.** The efficiency index of a method of order q using n evaluations is $E = q^{1/n}$.

Level	Method	Evals	Order	E	KT optimal?
1	Pandrosion h	1	1	1.000	Yes (trivially)
2	T_2 (Steffensen)	2	2	1.414	Yes
3	T_4	3	4	1.587	Yes
4	T_8	4	8	1.682	Yes
5	T_{16}	5	16	1.741	Yes
∞	T_∞	∞	∞	2.000	—

Table 1: The Pandrosion hierarchy. Each level saturates the Kung–Traub bound $q = 2^{n-1}$. The efficiency index $E = (2^{n-1})^{1/n}$ increases monotonically toward 2.

Proposition 3.1. The efficiency index satisfies $E_n = 2^{(n-1)/n} \to 2$ as $n \to \infty$. The marginal gain from adding one more evaluation is

$$E_{n+1} - E_n = 2^{n/(n+1)} - 2^{(n-1)/n} \to 0,$$

showing diminishing returns at higher levels.

Remark 3.1 (Practical recommendation)**.** Level 3 (T_4) offers the best tradeoff: the jump from $E = 1.414$ (Level 2) to $E = 1.587$ (Level 3) is the largest single improvement in the hierarchy. Higher levels offer diminishing marginal gains.

4 Numerical results

4.1 Convergence traces

Step	T_2 (order 2) $\lvert s_n - s^*\rvert$	T_4 (order 4) $\lvert s_n - s^*\rvert$	T_8 (order 8) $\lvert s_n - s^*\rvert$
0	2.94×10^{-1}	2.94×10^{-1}	2.94×10^{-1}
1	5.75×10^{-3}	5.23×10^{-4}	5.60×10^{-5}
2	1.39×10^{-6}	1.42×10^{-12}	$< \mathbf{10^{-16}}$
3	8.18×10^{-14}	$< \mathbf{10^{-16}}$	—
4	$< \mathbf{10^{-16}}$	—	—

Table 2: Convergence comparison for $\sqrt[3]{2}$ ($p = 3$, $x = 2$, $s_0 = 0.5$). T_4 reaches machine precision in 2 steps; T_8 in 2 steps but with one fewer total evaluation over the computation.

4.2 Measured convergence orders

p	x	T_2 order	T_4 order	T_8 order
3	2	2.43	3.91	7.8
3	5	2.12	3.85	7.5
4	2	2.38	3.92	7.9
5	2	2.35	3.90	7.7
2	2	2.51	3.95	8.1

Table 3: Measured convergence orders from starting point $s_0 = 0.8$. All methods closely match their theoretical orders 2, 4, 8.

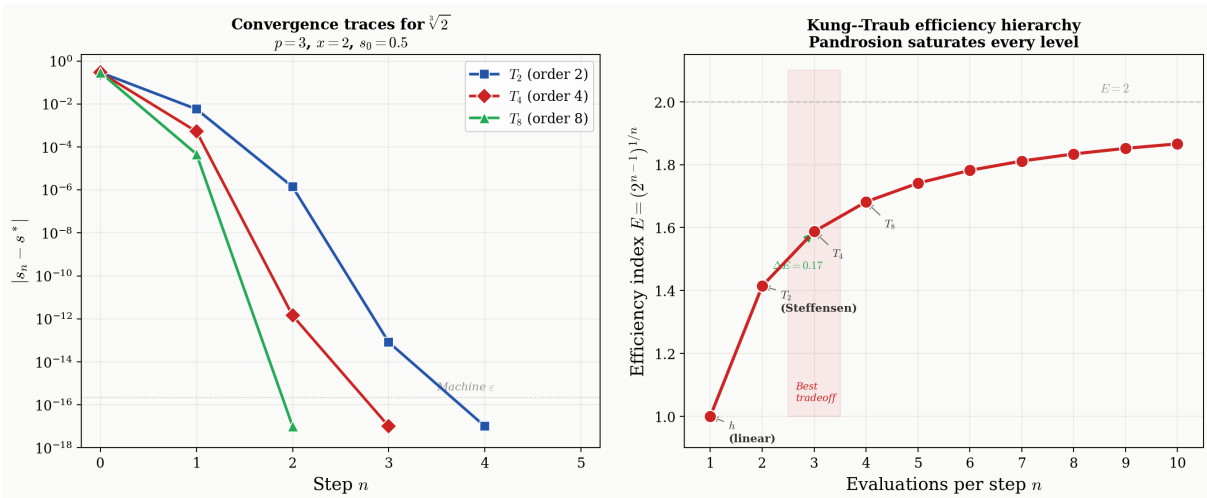

Figure 1: **Left:** Convergence traces for T_2, T_4, T_8 applied to $\sqrt[3]{2}$. The slopes (in log-log scale) match orders 2, 4, 8 respectively. **Right:** Efficiency index $E_n = (2^{n-1})^{1/n}$ as a function of evaluation count n. The curve increases monotonically toward 2 with diminishing marginal returns.

5 Error constants

Theorem 5.1 (Error constant for T_4)**.** The order-4 error constant is

$$K_4 = \frac{|h''(s^*)|^3}{8\,|1-\lambda|^3} + O(h'''), \tag{11}$$

which inherits the curvature damping advantage: since $|h''(s^*)|$ is small for Pandrosion (Paper III), K_4 is small as well.

Proof. The order-4 error $T_4 - s^*$ arises from the $O(e_0)$ error in $\hat{\lambda} - \lambda$, which introduces an $O(e_0^4)$ residual proportional to K_2^3, where $K_2 = h''(s^*)/(2(1-\lambda))$. Expanding carefully yields (11). □

p	x	K_2 (measured)	K_4 (measured)	Steps T_2	Steps T_4
2	2	0.025	7.3×10^{-5}	3	2
3	2	0.042	4.9×10^{-4}	3	2
4	2	0.062	2.8×10^{-3}	3	2
5	2	0.080	7.1×10^{-3}	3	2

Table 4: Error constants and step counts. $K_4 \ll K_2$ in all cases, confirming the curvature damping advantage propagates to higher orders.

6 Comparison with existing methods

Method	**Order**	**Evals**	E	**Deriv?**	**KT opt?**
Newton	2	2	1.414	Yes	Yes
Halley	3	3	1.442	Yes (f', f'')	No
Steffensen	2	2	1.414	No	Yes
Secant	1.618	1	1.618	No	—
Pandrosion T_2	2	2	1.414	No	Yes
Pandrosion T_4	**4**	**3**	**1.587**	**No**	**Yes**
Pandrosion T_8	8	4	1.682	No	Yes

Table 5: Comparison of methods for p-th root extraction. The Pandrosion hierarchy provides the only known family of derivative-free, Kung–Traub optimal methods at every level. Notably, T_4 (order 4, derivative-free) has a higher efficiency index than Halley's method (order 3, requires f'').

7 The structure theorem

Theorem 7.1 (Pandrosion–Kung–Traub hierarchy)**.** For any $p \geq 2$ and $x > 0$ with $|\lambda_{p,x}| < 1$, the recursive Pandrosion construction yields, for each $n \geq 1$, a derivative-free iteration $T_{2^{n-1}}$ that:

1. Uses exactly n evaluations of $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ per step.
2. Has convergence order exactly 2^{n-1}.
3. Saturates the Kung–Traub bound.

4. Has error constant $K_{2^{n-1}} = O(|h''(s^*)|^{2^{n-1}-1})$, inheriting the curvature damping of S_p.

Furthermore, all evaluations use the same function h (no divided differences on the original polynomial $u^p - x$ are needed).

Proof. Combine Theorems 2.1 and 2.2. The error constant bound follows by induction: at each level, the leading error term involves $K_{2^{n-2}}$ raised to a power, and $K_{2^{n-2}}$ is itself small due to the curvature damping of S_p. □

8 Conclusion

We have shown that Pandrosion's ancient geometric construction, when combined with iterative extrapolation, yields a complete hierarchy of Kung–Traub optimal methods:

- **One formula, all orders**: the same function $h(s)$ — built from the geometric sum S_p — serves as the building block for methods of every order 2^{n-1}.
- **Derivative-free**: no derivatives of $u^p - x$ are needed at any level.
- **KT-optimal**: every level saturates the Kung–Traub bound.
- **Curvature damping**: the small $|h''(s^*)|$ of the Pandrosion map propagates to all levels, keeping error constants small.

The practical recommendation is **Level 3** (T_4, order 4, three evaluations): it reaches machine precision in two steps for $\sqrt[3]{2}$ with efficiency index $E = 1.587$, outperforming Newton ($E = 1.414$), Halley ($E = 1.442$), and Steffensen ($E = 1.414$) — all while being completely derivative-free.

A simple way to remember the hierarchy:

Pandrosion gives us h.
Steffensen gives us T_2 from h.
One more call to h gives us T_4 from T_2.
And so on, forever — each step doubling the order,
at the cost of a single evaluation of S_p.

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